

DMVPN Fase 2 — IKEv1 con Enrutamiento Dinámico (EIGRP)

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Plataforma: Lab PNET

Video demostrativo: https://www.youtube.com/watch?v=yKM_9PRwaSc

Enlace GitHub: <https://github.com/juniorjaversantosperez/DMVPN-Fase-2-con-IKEv1-con-enrutamiento-dinamico.git>

Objetivo

Implementar una VPN hub-and-spoke punto a multipunto usando **DMVPN Fase 2** con **IKEv1** como protocolo de negociación de claves y **EIGRP** como protocolo de enrutamiento dinámico.

La Fase 2 de DMVPN permite que los Spokes establezcan túneles **directos entre sí** (Spoke-to-Spoke) sin necesidad de que el tráfico pase por el Hub, mejorando la latencia y reduciendo la carga en el nodo central. IKEv1 gestiona la negociación de las asociaciones de seguridad en dos fases (Main Mode + Quick Mode).

Topología

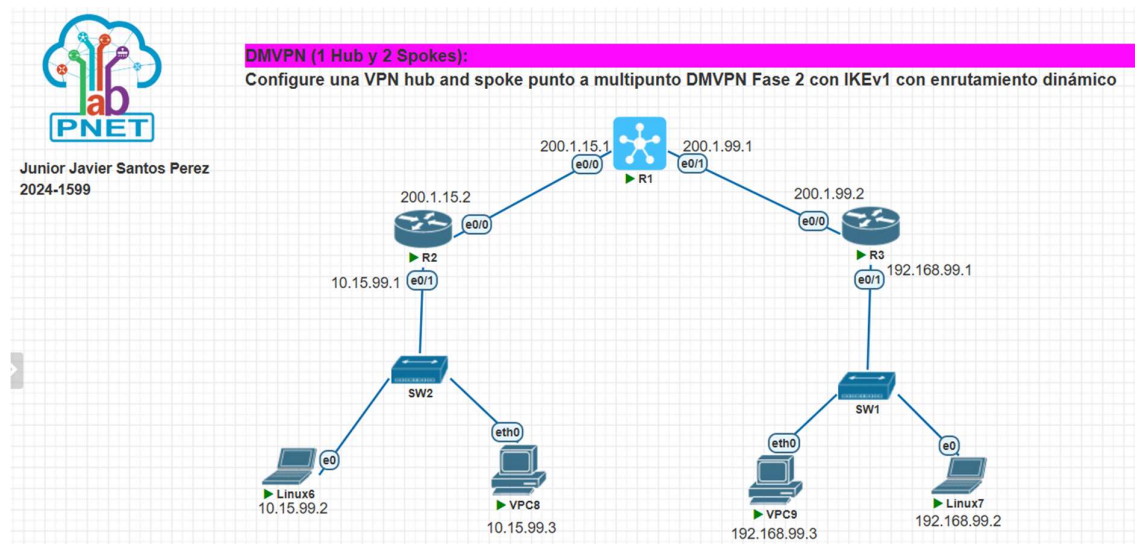


IMAGEN1 — Topología general del laboratorio: R1 como Hub central, R2 y R3 como Spokes, con sus respectivas LANs conectadas a través de SW1 y SW2.

Dispositivos y roles

Dispositivo	Rol	Descripción
R1	Hub	Nodo central del DMVPN
R2	Spoke 1	Sitio izquierdo (LAN 10.15.99.0)
R3	Spoke 2	Sitio derecho (LAN 192.168.99.0)
SW2	Switch	Conmutador LAN del Spoke 1
SW1	Switch	Conmutador LAN del Spoke 2

Direccionamiento IP

Interfaces físicas (NBMA)

Dispositivo	Interfaz	Dirección IP	Red
R1 Hub	e0/0	200.1.15.1/24	Red hacia R2
R1 Hub	e0/1	200.1.99.1/24	Red hacia R3
R2 Spoke1	e0/0	200.1.15.2/24	Red NBMA
R2 Spoke1	e0/1	10.15.99.1/24	LAN Spoke1
R3 Spoke2	e0/0	200.1.99.2/24	Red NBMA
R3 Spoke2	e0/1	192.168.99.1/24	LAN Spoke2

Interfaz de túnel mGRE

Dispositivo	Interfaz	IP Túnel
R1 Hub	Tunnel10	172.16.0.1/24
R2 Spoke1	Tunnel10	172.16.0.2/24
R3 Spoke2	Tunnel10	172.16.0.3/24

Hosts

Host	Interfaz	IP	Gateway
Linux6	e0	10.15.99.2/24	10.15.99.1
VPC8	eth0	10.15.99.3/24	10.15.99.1
Linux7	e0	192.168.99.2/24	192.168.99.1
VPC9	eth0	192.168.99.3/24	192.168.99.1

Parámetros de configuración

IKEv1 — Fase 1 (ISAKMP)

Parámetro	Valor
Cifrado	AES
Hash	SHA
Autenticación	Pre-shared
Grupo DH	2
Lifetime	86400 seg
PSK	cisco123

IPSec — Fase 2

Parámetro	Valor
Transform-set	DMVPN-SET
Encriptación ESP	AES
Autenticación ESP	SHA-HMAC
Modo	Transport
Perfil IPSec	DMVPN-PROF

NHRP

Parámetro	Valor
Network-ID	100
Autenticación	dmvpn1
NHS (Hub IP túnel)	172.16.0.1
NHS (Hub NBMA)	200.1.15.1

EIGRP

Parámetro	Valor
AS Number	100
Auto-summary	Off

Configuración completa

Paso 1 — Interfaces físicas

R1 Hub:

```
interface e0/0
  ip address 200.1.15.1 255.255.255.0
  no shut
interface e0/1
  ip address 200.1.99.1 255.255.255.0
  no shut
```

R2 Spoke1:

```
interface e0/0
  ip address 200.1.15.2 255.255.255.0
  no shut
interface e0/1
  ip address 10.15.99.1 255.255.255.0
```

```
no shut
ip route 0.0.0.0 0.0.0.0 200.1.15.1
```

R3 Spoke2:

```
interface e0/0
ip address 200.1.99.2 255.255.255.0
no shut
interface e0/1
ip address 192.168.99.1 255.255.255.0
no shut
ip route 0.0.0.0 0.0.0.0 200.1.99.1
```

Paso 2 — IKEv1 Fase 1 (ISAKMP)

El Hub usa address 0.0.0.0 0.0.0.0 para aceptar conexiones de cualquier Spoke dinámicamente. Los Spokes apuntan a la IP NBMA del Hub.

R1 Hub:

```
crypto isakmp policy 10
encr aes
hash sha
authentication pre-share
group 2
lifetime 86400

crypto isakmp key cisco123 address 0.0.0.0 0.0.0.0
```

R2 Spoke1 y R3 Spoke2:

```
crypto isakmp policy 10
encr aes
hash sha
authentication pre-share
```

```
group 2
lifetime 86400
```

```
crypto isakmp key cisco123 address 200.1.15.1
```

Paso 3 — IPSec Fase 2

Modo transport porque el encapsulamiento GRE ya es realizado por la interfaz mGRE.

R1, R2 y R3 (igual en todos):

```
crypto ipsec transform-set DMVPN-SET esp-aes esp-sha-hmac
mode transport
```

```
crypto ipsec profile DMVPN-PROF
set transform-set DMVPN-SET
```

Paso 4 — Interfaz Tunnel10 mGRE + NHRP

En Fase 2, el Hub **no** usa `ip nhrp redirect` y los Spokes **no** usan `ip nhrp shortcut`.

El Hub no debe resumir rutas sobre Tunnel10 para que los Spokes vean el next-hop real del destino.

R1 Hub:

```
interface Tunnel10
ip address 172.16.0.1 255.255.255.0
no ip redirects
ip nhrp network-id 100
ip nhrp map multicast dynamic
ip nhrp authentication dmvpn1
tunnel source e0/0
tunnel mode gre multipoint
```

```
tunnel protection ipsec profile DMVPN-PROF
```

R2 Spoke1:

```
interface Tunnel10
 ip address 172.16.0.2 255.255.255.0
 no ip redirects
 ip nhrp network-id 100
 ip nhrp authentication dmvpn1
 ip nhrp nhs 172.16.0.1
 ip nhrp map 172.16.0.1 200.1.15.1
 ip nhrp map multicast 200.1.15.1
 ip nhrp registration no-unique
 tunnel source e0/0
 tunnel mode gre multipoint
 tunnel protection ipsec profile DMVPN-PROF
```

R3 Spoke2:

```
interface Tunnel10
 ip address 172.16.0.3 255.255.255.0
 no ip redirects
 ip nhrp network-id 100
 ip nhrp authentication dmvpn1
 ip nhrp nhs 172.16.0.1
 ip nhrp map 172.16.0.1 200.1.15.1
 ip nhrp map multicast 200.1.15.1
 ip nhrp registration no-unique
 tunnel source e0/0
 tunnel mode gre multipoint
 tunnel protection ipsec profile DMVPN-PROF
```

Paso 5 — EIGRP dinámico

`no ip split-horizon eigrp 100` es obligatorio en el Hub para que pueda reenviar rutas aprendidas de un Spoke hacia otros Spokes por la misma interfaz Tunnel10.

R1 Hub:

```
router eigrp 100
  network 172.16.0.0 0.0.0.255
  network 10.15.99.0 0.0.0.255
  network 192.168.99.0 0.0.0.255
  no auto-summary

interface Tunnel10
  no ip split-horizon eigrp 100
  no ip summary-address eigrp 100 172.16.0.0 255.255.255.0
```

R2 Spoke1:

```
router eigrp 100
  network 172.16.0.0 0.0.0.255
  network 10.15.99.0 0.0.0.255
  no auto-summary
```

R3 Spoke2:

```
router eigrp 100
  network 172.16.0.0 0.0.0.255
  network 192.168.99.0 0.0.0.255
  no auto-summary
```


Paso 6 — Hosts

Linux6:

```
ip addr add 10.15.99.2/24 dev eth0
ip route add default via 10.15.99.1
```

Linux7:

```
ip addr add 192.168.99.2/24 dev eth0
ip route add default via 192.168.99.1
```

VPC8:

```
ip 10.15.99.3 255.255.255.0 10.15.99.1
```

VPC9:

```
ip 192.168.99.3 255.255.255.0 192.168.99.1
```

Demostración del funcionamiento

Estado DMVPN en R1 Hub

```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
R1-HUB#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
         N - NATed, L - Local, X - No Socket
         T1 - Route Installed, T2 - Nexthop-override
         C - CTS Capable, I2 - Temporary
         # Ent --> Number of NHRP entries with same NBMA peer
         NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
         UpDn Time --> Up or Down Time for a Tunnel
=====
Interface: Tunnel10, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
  1 200.1.15.2          172.16.0.2    UP 00:30:34    D
  1 200.1.99.2          172.16.0.3    UP 00:27:15    D
```

IMAGEN2 — `show dmvpn` en R1-HUB. Se observan 2 peers dinámicos (atributo D) con estado UP:

- 200.1.15.2 → 172.16.0.2 (R2 Spoke1) — activo 00:30:34
- 200.1.99.2 → 172.16.0.3 (R3 Spoke2) — activo 00:27:15

Ambos Spokes están registrados correctamente en el Hub a través de NHRP.

Estado IKEv1 en R1 Hub

```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
R1-HUB#sho
R1-HUB#show cry
R1-HUB#show crypto is
R1-HUB#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
200.1.15.1   200.1.99.2   QM_IDLE       1002 ACTIVE
200.1.15.1   200.1.15.2   QM_IDLE       1001 ACTIVE

IPv6 Crypto ISAKMP SA
```

IMAGEN3 — `show crypto isakmp sa` en R1-HUB. Se muestran dos sesiones IKEv1 en estado QM_IDLE (Quick Mode Idle = activo y estable):

- 200.1.99.2 → 200.1.15.1 — conn-id 1002 ACTIVE (R3 al Hub)
- 200.1.15.2 → 200.1.15.1 — conn-id 1001 ACTIVE (R2 al Hub)

Vecinos EIGRP en R1 Hub

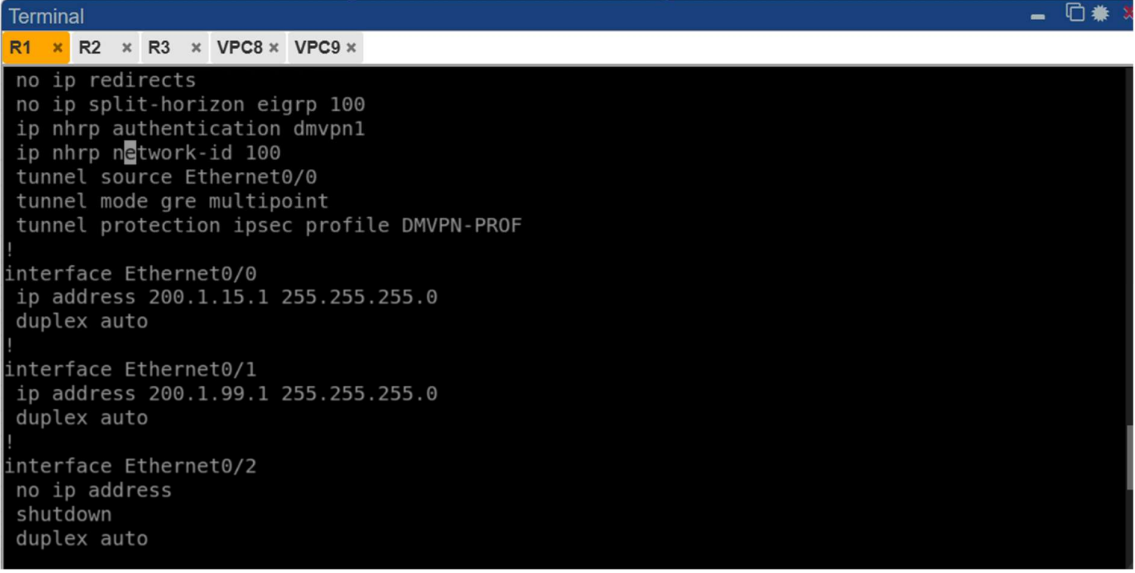
```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
IPv6 Crypto ISAKMP SA

R1-HUB#sho
R1-HUB#show ip ei
R1-HUB#show ip eigrp ne
R1-HUB#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address          Interface          Hold Uptime      SRTT   RT0  Q  Seq
                               (sec)              (ms)
1   172.16.0.3         Tu10               13 00:22:28      49  1434 0  4
0   172.16.0.2         Tu10               11 00:23:46      48  1434 0  4
R1-HUB#
```

IMAGEN4 — show ip eigrp neighbors en R1-HUB. EIGRP AS 100 ha formado adyacencias con ambos Spokes sobre Tunnel10:

- 172.16.0.3 (R3) — uptime 00:22:28
- 172.16.0.2 (R2) — uptime 00:23:46

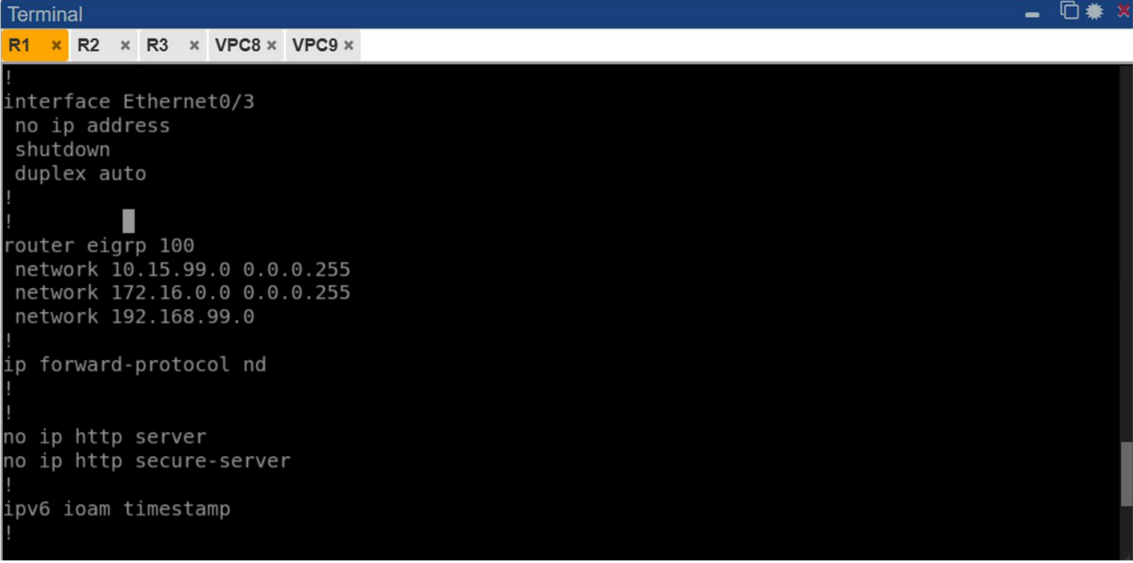
Configuración Tunnel10 e interfaces en R1 Hub



```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
no ip redirects
no ip split-horizon eigrp 100
ip nhrp authentication dmvpn1
ip nhrp network-id 100
tunnel source Ethernet0/0
tunnel mode gre multipoint
tunnel protection ipsec profile DMVPN-PROF
!
interface Ethernet0/0
ip address 200.1.15.1 255.255.255.0
duplex auto
!
interface Ethernet0/1
ip address 200.1.99.1 255.255.255.0
duplex auto
!
interface Ethernet0/2
no ip address
shutdown
duplex auto
```

IMAGEN5 — Extracto de show run en R1-HUB mostrando la configuración de Tunnel10: no ip redirects, autenticación NHRP, tunnel mode gre multipoint y protección IPSec con DMVPN-PROF. Se confirman también las IPs de e0/0 (200.1.15.1) y e0/1 (200.1.99.1).

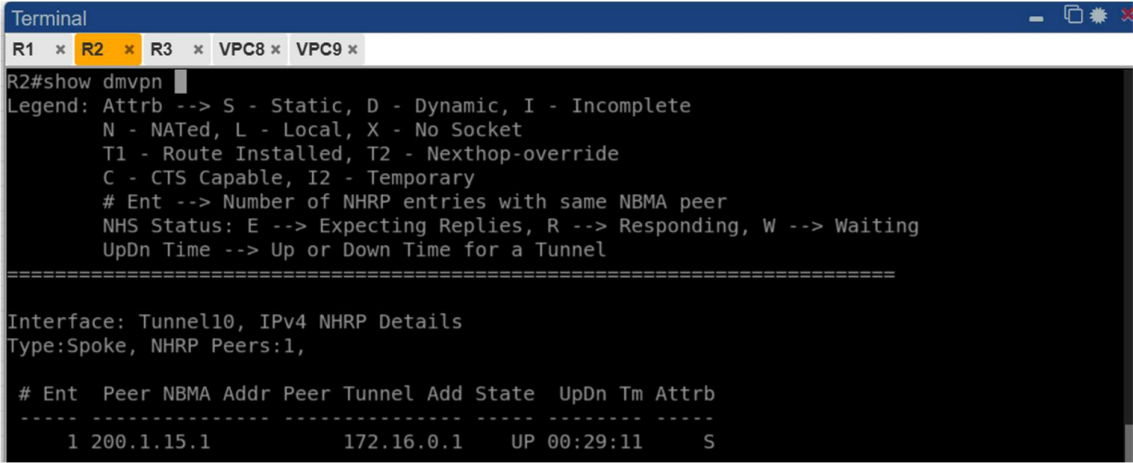
EIGRP y rutas en R1 Hub

A terminal window titled "Terminal" with tabs for R1, R2, R3, VPC8, and VPC9. The R1 tab is active. The terminal shows the configuration of interface Ethernet0/3 and the EIGRP 100 process. The configuration includes setting the interface to no ip address, shutdown, and duplex auto. The EIGRP process is configured with three networks: 10.15.99.0, 172.16.0.0, and 192.168.99.0. Other settings include ip forward-protocol nd, no ip http server, no ip http secure-server, and ipv6 ioam timestamp.

```
!
interface Ethernet0/3
no ip address
shutdown
duplex auto
!
!
router eigrp 100
network 10.15.99.0 0.0.0.255
network 172.16.0.0 0.0.0.255
network 192.168.99.0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
!
```

IMAGEN6 — Extracto de `show run` en R1-HUB mostrando la configuración de router eigrp 100 con las tres redes anunciadas: 10.15.99.0, 172.16.0.0 y 192.168.99.0.

Estado DMVPN en R2 Spoke1

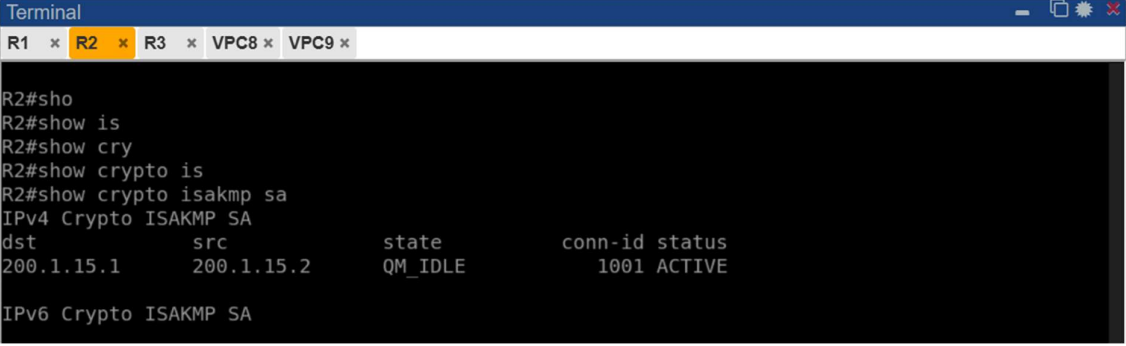
A terminal window titled "Terminal" with tabs for R1, R2, R3, VPC8, and VPC9. The R2 tab is active. The terminal shows the output of the command "show dmvpn". It includes a legend for attributes (S for Static, D for Dynamic, I for Incomplete, N for NATed, L for Local, X for No Socket, T1 for Route Installed, T2 for Nexthop-override, C for CTS Capable, I2 for Temporary) and NHS status (E for Expecting Replies, R for Responding, W for Waiting). The output shows details for Tunnel10, which is a Spoke with 1 NHRP peer. The peer is 200.1.15.1, with a next-hop address of 172.16.0.1, and is in the UP state with a static attribute (S).

```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
R2#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====
Interface: Tunnel10, IPv4 NHRP Details
Type:Spoke, NHRP Peers:1,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
1 200.1.15.1          172.16.0.1    UP 00:29:11    S
```

IMAGEN7 — `show dmvpn` en R2. Tipo: Spoke con 1 peer. Muestra el Hub 200.1.15.1 → 172.16.0.1 con estado UP (atributo S = Static, entrada estática configurada en el Spoke).

Estado IKEv1 en R2 Spoke1



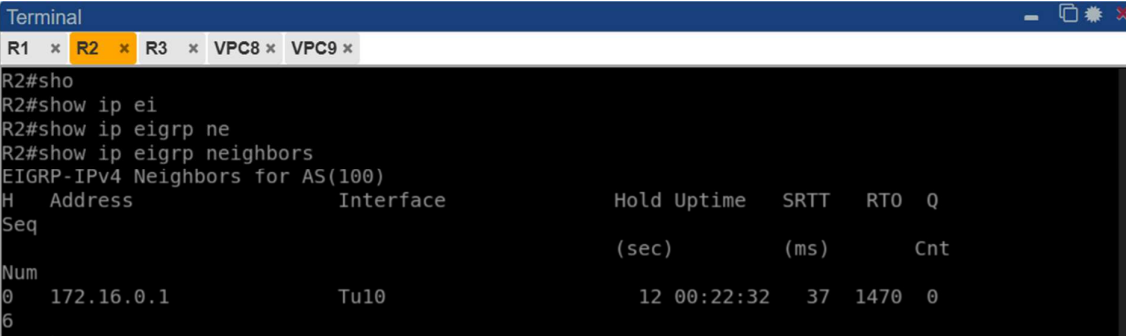
```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x

R2#sho
R2#show is
R2#show cry
R2#show crypto is
R2#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
200.1.15.1   200.1.15.2   QM_IDLE       1001  ACTIVE

IPv6 Crypto ISAKMP SA
```

IMAGEN8 — `show crypto isakmp sa` en R2. Sesión IKEv1 en estado `QM_IDLE` `ACTIVE`: `200.1.15.2` → `200.1.15.1` (R2 hacia el Hub). La negociación IKEv1 completada exitosamente.

Vecinos EIGRP en R2 Spoke1



```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x

R2#sho
R2#show ip ei
R2#show ip eigrp ne
R2#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address          Interface          Hold Uptime    SRTT   RT0    Q
Seq                                     (sec)         (ms)    Cnt
Num
0   172.16.0.1         Tu10              12 00:22:32   37  1470   0
6
```

IMAGEN9 — `show ip eigrp neighbors` en R2. EIGRP AS 100 ha formado adyacencia con el Hub `172.16.0.1` sobre `Tu10` (Tunnel10) con uptime `00:22:32`.

Estado DMVPN en R3 Spoke2

```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
R3#show dm
R3#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
=====
Interface: Tunnel10, IPv4 NHRP Details
Type:Spoke, NHRP Peers:1,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
  1 200.1.15.1          172.16.0.1   UP 00:28:12    S
```

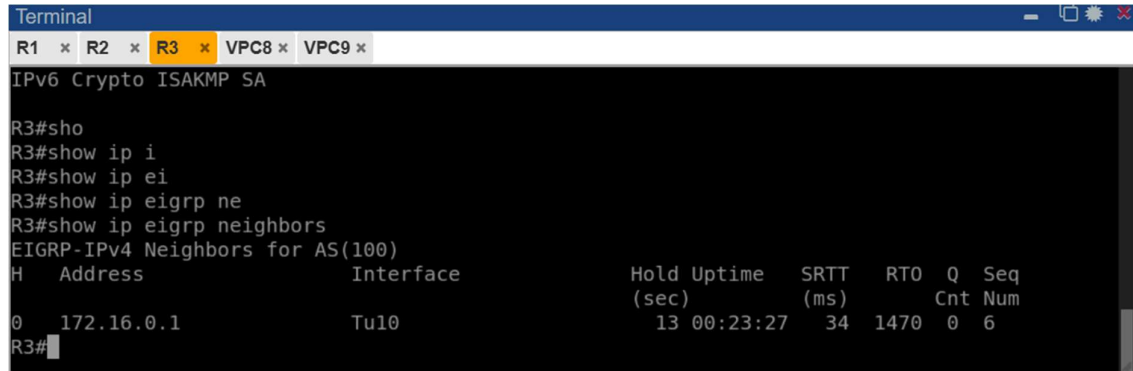
IMAGEN10 — `show dmvpn` en R3. Tipo: Spoke con 1 peer. Hub 200.1.15.1 → 172.16.0.1 con estado UP (atributo S). R3 alcanza al Hub a través de su gateway 200.1.99.1.

Estado IKEv1 en R3 Spoke2

```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
R3#sho
R3#show cry
R3#show crypto is
R3#show crypto isakmp s
R3#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
200.1.15.1   200.1.99.2   QM_IDLE       1001 ACTIVE
IPv6 Crypto ISAKMP SA
```

IMAGEN11 — `show crypto isakmp sa` en R3. Sesión IKEv1 QM_IDLE ACTIVE: 200.1.99.2 → 200.1.15.1. R3 se autentica con la IP NBMA del Hub (200.1.15.1), que es la fuente del Tunnel10.

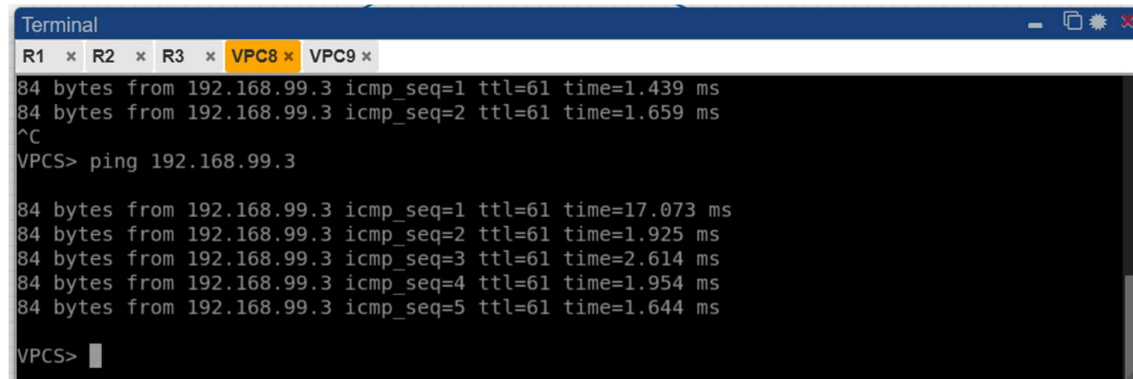
Vecinos EIGRP en R3 Spoke2



```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
IPv6 Crypto ISAKMP SA
R3#sho
R3#show ip i
R3#show ip ei
R3#show ip eigrp ne
R3#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address                  Interface      Hold Uptime    SRTT   RT0  Q  Seq
                               (sec)          (ms)          Cnt  Num
0   172.16.0.1                Tu10          13 00:23:27   34  1470  0   6
R3#
```

IMAGEN12 — *show ip eigrp neighbors en R3. EIGRP AS 100 formó adyacencia con el Hub 172.16.0.1 sobre Tu10 con uptime 00:23:27.*

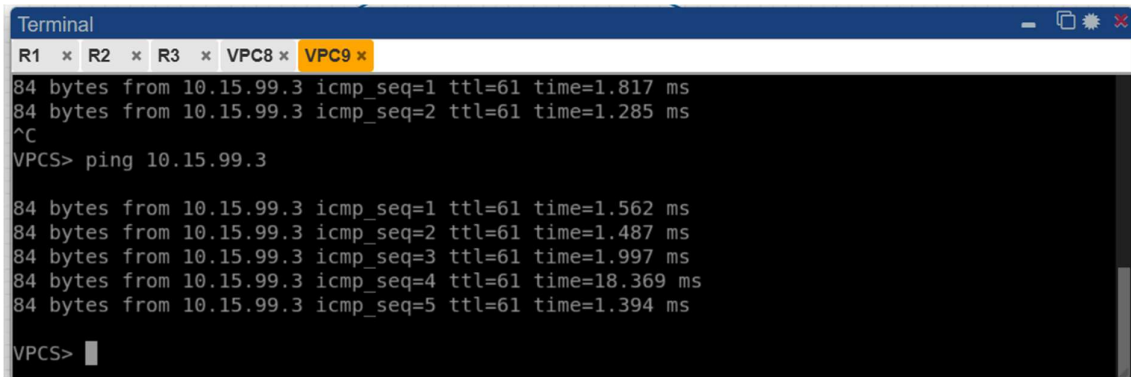
Ping VPC8 → VPC9



```
Terminal
R1 x R2 x R3 x VPC8 x VPC9 x
84 bytes from 192.168.99.3 icmp_seq=1 ttl=61 time=1.439 ms
84 bytes from 192.168.99.3 icmp_seq=2 ttl=61 time=1.659 ms
^C
VPCS> ping 192.168.99.3
84 bytes from 192.168.99.3 icmp_seq=1 ttl=61 time=17.073 ms
84 bytes from 192.168.99.3 icmp_seq=2 ttl=61 time=1.925 ms
84 bytes from 192.168.99.3 icmp_seq=3 ttl=61 time=2.614 ms
84 bytes from 192.168.99.3 icmp_seq=4 ttl=61 time=1.954 ms
84 bytes from 192.168.99.3 icmp_seq=5 ttl=61 time=1.644 ms
VPCS>
```

IMAGEN13 — *Ping desde VPC8 (10.15.99.3) hacia VPC9 (192.168.99.3). Resultado: 5/5 paquetes exitosos, latencia promedio ~2ms. El tráfico viaja cifrado por el túnel DMVPN entre Spoke1 (R2) y Spoke2 (R3).*

Ping VPC9 → VPC8



*IMAGEN14 — Ping desde VPC9 (192.168.99.3) hacia VPC8 (10.15.99.3).
Resultado: 5/5 paquetes exitosos. Se confirma la conectividad bidireccional entre ambas LANs a través del DMVPN Fase 2 con IKEv1.*

Resumen de verificación

Verificación	Comando	Resultado esperado	Estado
Túneles NHRP Hub	show dmvpn	2 peers UP (D = Dynamic)	✓
Sesiones IKEv1 Hub	show crypto isakmp sa	QM_IDLE ACTIVE (x2)	✓
Vecinos EIGRP Hub	show ip eigrp neighbors	172.16.0.2 y 172.16.0.3 en Tu10	✓
Túnel NHRP R2	show dmvpn	Hub UP (S = Static)	✓
IKEv1 R2	show crypto isakmp sa	QM_IDLE ACTIVE	✓
EIGRP R2	show ip eigrp neighbors	172.16.0.1 en Tu10	✓
Túnel NHRP R3	show dmvpn	Hub UP (S = Static)	✓
IKEv1 R3	show crypto isakmp sa	QM_IDLE ACTIVE	✓

Verificación	Comando	Resultado esperado	Estado
EIGRP R3	show ip eigrp neighbors	172.16.0.1 en Tu10	✓
Conectividad VPC8 → VPC9	ping 192.168.99.3	5/5 exitosos	✓
Conectividad VPC9 → VPC8	ping 10.15.99.3	5/5 exitosos	✓

Notas técnicas

- Los túneles Spoke-to-Spoke en Fase 2 se crean **bajo demanda** al primer paquete de tráfico. El primer ping puede mostrar latencia alta (~17ms) por la negociación inicial; los siguientes son normales (~2ms).
- El Hub **no debe resumir** rutas sobre Tunnel10 en Fase 2: los Spokes necesitan ver el next-hop real del Spoke destino para construir el túnel directo.
- `no ip split-horizon eigrp 100` en Tunnel10 del Hub es obligatorio para que EIGRP propague rutas entre Spokes.
- R3 usa 200.1.99.1 como gateway pero se registra en NHRP con la IP 200.1.15.1 (fuente del Tunnel10 del Hub), ya que R1 enruta internamente entre sus dos interfaces WAN.